## CONTROL SYSTEM DEVICE AND CONTROL RELATIONSHIP CHART [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

id0103q1800200

×: Applicable

| Item | MAIN RELAY CONTROL | DRIVE-BY-WIRE CONTROL | HYDRAULIC VARIABLE VALVE TIMING CONTROL | ELECTRIC VARIABLE VALVE TIMING CONTROL | FUEL INJECTION CONTROL | FUEL PUMP CONTROL | HIGH PRESSURE FUEL PUMP CONTROL | ELECTRONIC SPARK ADVANCE CONTROL | PURGE CONTROL | A/F SENSOR HEATER CONTROL HO2S HEATER CONTROL | A/C CUT-OFF CONTROL | ELECTRICAL FAN CONTROL | STARTER CUT-OFF CONTROL | GENERATOR CONTROL | i-stop CONTROL | DC-DC CONVERTER CONTROL | ENGINE OIL CONTROL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Input device | | | | | | | | | | | | | | | | | |
| APP sensor No.1, No.2 | | × | × | | × | | | × | × | | × | × | | | × | | |
| TP sensor No.1, No.2 | | × | | | × | | | | | | × | | | | × | | |
| MAF sensor | | × | × | × | × | | × | × | × | × | | × | | | × | | × |
| IAT sensor No.1 | | × | | × | × | | | × | × | | | × | | ×*3 | × | | |
| IAT sensor No.2 | | × | | × | × | | | × | | | | | | | × | | |
| MAP sensor | | × | | × | × | | | × | × | | | | | | × | | × |
| CKP sensor | | × | × | × | × | × | × | × | × | × | × | × | × | × | × | | × |
| Intake CMP sensor | | × | ×*5 | × | × | | | × | | | | | | | × | | |
| Exhaust CMP sensor*6 | | × | × | | × | | × | × | | | | | | | × | | |
| ECT sensor | | × | × | × | × | | × | × | × | × | × | × | × | ×*3 | × | | × |
| Fuel pressure sensor | | × | | | × | × | × | | | | | | | | × | | |
| BARO sensor | | × | | × | × | | | | × | | | × | | | × | | |
| Power brake unit vacuum sensor*2 | | | | | | | | | | | | | | | × | | |
| Ion sensor No.1—No.4*6 | | | | × | × | | | | | | | | | | | | |
| KS | | | | | | | | × | | | | | | | | | |
| A/F sensor | | | | | × | | | | × | | | | | | | | |
| HO2S | | | | | × | | | | | | | | | | | | |
| Refrigerant pressure sensor*7 | | | | | | | | | | | × | × | | | | | |
| IG1 relay | × | | × | | × | × | | × | | × | | × | × | | × | × | |
| Brake switch (No.1 signal) | | × | | | | | | | | | | | | | × | | |
| Brake switch (No.2 signal) | | × | | | | | | | | | | | | | | | |
| Transaxle range sensor (ATX) | | × | | | × | | | | | | | | × | | × | | |
| CPP switch (MTX) | | × | | | × | | | × | | | × | | | | × | | |
| Starter interlock switch (MTX) | | | | | | | | | | | | | × | | | | |
| Clutch stroke sensor (MTX) *2 | | | | | | | | | | | | | | | × | | |
| Neutral switch No.1 (MTX) | | × | | | × | | | × | | | × | | | | × | | |
| Neutral switch No.2 (MTX)*2 | | | | | | | | | | | | | | | × | | |
| Electric variable valve timing motor/driver*6 | | | | × | | | | | | | | | | | × | | |
| Battery | | | × | | | × | × | × | × | | | | | × | × | | |
| Generator (Stator coil) | | × | | | | | | | | | | | | | | | |
| CAN*1 | | × | | | × | | | × | | | × | × | × | ×*2 | × | × | |
| Output device | | | | | | | | | | | | | | | | | |
| Main relay | × | | | | | | | | | | | | | | | | |
| Throttle valve actuator | | × | | | | | | | | | | | | | × | | |
| Electric variable valve timing motor/driver*6 | | | | × | × | | | | | | | | | | × | | |
| Electric variable valve timing relay*6 | | | | × | | | | | | | | | | | × | | |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| OCV | | | × | | | | | | | | | | | | | | |
| Fuel injector No.1—No.4 | | | | | × | | | | | | | | | | × | | |
| Fuel injector relay | | | | | × | | | | | | | | | | × | | |
| Fuel pump control module*8 | | | | | | × | | | | | | | | | | | |
| Fuel pump relay | | | | | | × | | | | | | | | | | | |
| Spill valve control solenoid valve (High pressure fuel pump) | | | | | | | × | | | | | | | | | | |
| Ignition coil No.1—No.4 | | | | | | | | × | | | | | | | × | | |
| Purge solenoid valve | | | | | | | | | × | | | | | | | | |
| A/F sensor heater | | | | | | | | | | × | | | | | | | |
| HO2S heater | | | | | | | | | | × | | | | | | | |
| A/C relay*7 | | | | | | | | | | | × | | | | | | |
| Cooling fan relay No.1, No.2 | | | | | | | | | | | | × | | | | | |
| Starter relay | | | | | | | | | | | | | × | | × | | |
| Generator (Field coil) | | | | | | | | | | | | | | × | × | | |
| DC-DC converter*4 | | | | | | | | | | | | | | | × | × | |
| Engine oil solenoid valve*6 | | | | | | | | | | | | | | | | | × |

*1 : TCM (ATX), DSC HU/CM or ABS HU/CM, EPS control module, instrument cluster, BCM, start stop unit
*2 : With i-stop system
*3 : Without i-stop system
*4 : With i-stop system, without i-ELOOP
*5 : SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1exhaust system)
*6 : SKYACTIV-G 1.5 (with 4-2-1exhaust system)
*7 : With air conditioner
*8 : With fuel pump control module